

The vertical normal forces follow immediately from the weights: the contact between A and the table carries $3mg$, between A and B carries $2mg$, and between B and C carries mg .

Q1. Slow push

(a) To make the whole stack slide as a rigid body, the applied force must overcome the maximum static friction between the bottom coin and the table. This contact carries the full weight $3mg$, so

$$\boxed{F = 3\mu_s mg}.$$

(b) Once the stack is sliding, the table exerts kinetic friction $3\mu_k mg$ on A . If the three coins still move together with common acceleration a , Newton's law for the whole stack gives

$$3m a = F - 3\mu_k mg \quad \implies \quad a = \frac{F - 3\mu_k mg}{3m}.$$

The friction between A and B must accelerate the two coins above it, so

$$f_{AB} = 2m a = \frac{2}{3}(F - 3\mu_k mg).$$

A begins to slip relative to B when this required force reaches the static limit $2\mu_s mg$:

$$\frac{2}{3}(F_{\text{crit}} - 3\mu_k mg) = 2\mu_s mg \implies \boxed{F_{\text{crit}} = 3(\mu_s + \mu_k) mg}.$$

At this force the B – C interface is also at its limit ($f_{BC} = ma = \mu_s mg$); since $\mu_s > \mu_k$, once A – B slips to kinetic friction the acceleration of B, C drops, so B and C remain stuck together, as stated.

Q2. Constant push

(a) With A – B slipping, the kinetic friction that A exerts on B is $2\mu_k mg$. This is the only horizontal force accelerating the combined $B + C$ (mass $2m$), so

$$2m a_B = 2\mu_k mg \implies \boxed{a_B = \mu_k g}.$$

(b) On coin A the forces are F forward, and the kinetic frictions $3\mu_k mg$ (table) and $2\mu_k mg$ (from B) backward, so

$$m a_A = F - 3\mu_k mg - 2\mu_k mg = F - 5\mu_k mg.$$

With $F = 5\mu_k mg$ this gives $a_A = 0$. The relative acceleration between B and A is therefore $\mu_k g$, and the relative displacement grows as

$$2R = \frac{1}{2} a_{\text{rel}} t^2 = \frac{1}{2} \mu_k g t^2 \implies \boxed{t = 2\sqrt{\frac{R}{\mu_k g}}}.$$

(c) With $F = 3\mu_k mg$, the force is less than the maximum static friction $3\mu_s mg$ that the table can supply (recall $\mu_s > \mu_k$). The stack therefore does not slide at all: there is **no** relative slip between A and B , and the whole system remains at rest.

Q3. Up the incline

(a) The whole process starts and ends at rest, so the total change in kinetic energy is zero. The only forces doing work are the push F (positive work W_p), the friction forces (total work W , negative), and gravity. Going up the incline, the three coins (total mass $3m$) rise a height $s \sin \theta$, so gravity does work $-3mg s \sin \theta$. The work-energy theorem gives

$$W_p + W - 3mg s \sin \theta = 0 \implies \boxed{W = 3mg s \sin \theta - W_p}.$$

(b) On the incline the forces opposing the motion are the down-slope component of gravity, $3mg \sin \theta$, and kinetic friction $3\mu_k mg \cos \theta$. Hence the deceleration has magnitude

$$\boxed{a = g(\sin \theta + \mu_k \cos \theta)}.$$

Since the stack stops after distance s from speed v_0 , the constant-acceleration relation $v_0^2 = 2as$ gives

$$\boxed{v_0 = \sqrt{2gs(\sin \theta + \mu_k \cos \theta)}}.$$